

nected to orange wires) bathed in an oscillating electric field. The light dislodges electrons, which accelerate in the electric field and emit waves of THz radiation. Fig. 3 shows the close-up of the electron source.

General applications

THz radiations are non-ionising, and therefore safe to humans. These penetrate a wide variety of non-conducting materials, including clothing, paper, plastics and ceramics, and can also penetrate fog and clouds, but are strongly absorbed by metal and water. Until recently, researchers did not extensively explore the material interactions occurring in the THz-spectral region because they lacked reliable sources of THz radiation. However, pressure to develop new THz sources arose from two dramatically different groups—ultra-fast time-domain spectroscopists who wanted to work with longer wavelengths, and long-wavelength radio astronomers who wanted to work with shorter wavelengths. Today, with continuous wave (CW) and pulsed sources readily available, investigators are pursuing potential THz-wavelength applications in many fields.

Companies are planning to exploit the commercial applications of TE. THz applications span the physical (security imaging), biological (cell formation) and medical (cancerous-tumour detection) sciences with a growing interest in the application of THz frequencies from security imaging through clothing in airport scanners to non-destructive pharmaceutical and manufacturing inspection through multilayered or opaque surfaces. The unique properties of THz radiation also include high-frequency radars to produce high-resolution images of objects through cloud, fog and dust storms to support aircraft landing in harsh environments.

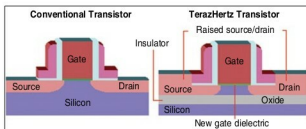


Fig. 4: Terahertz transistor

Much of the recent interest in THz radiation stems from its ability to penetrate deep into many organic materials without the damage associated with ionising radiation such as X-rays. Also, because THz radiation is readily absorbed by water, it can be used to distinguish between materials with varying water content, for example, fat versus lean meat. These properties lend themselves to applications in process and quality control as well as biomedical imaging. Tests are currently underway to determine whether THz tomographic imaging can augment or replace mammography, and some people have proposed THz imaging as a method of screening passengers for explosives at airports as well as for detecting the presence of cancerous cells in humans. However, all these applications are still in the research phase.

THz radiation can also help scientists understand the complex dynamics involved in condensed-matter physics and processes such as molecular recognition and protein folding. CW THz technology has long interested astronomers because approximately one-half of the total luminosity and 98 per cent of the photons emitted since the Big Bang fall into the sub-millimetre and far-IR, and CW THz sources can be used to help study these photons. One type of CW THz source is the optically pumped THz laser (OPTL). These lasers are in use around the world, primarily for astronomy, environmental monitoring and plasma diagnostics. The emerging field of time-domain spectroscopy (TDS) typically relies on a broadband short-pulse THz source. A split an-

tenna is fabricated on a semiconductor substrate to create a switch. A DC bias is placed across the antenna, and an ultra-short pump-laser pulse (< 100fs) is focused in the gap in the

antenna. The bias-laser pulse combination allows electrons to rapidly jump the gap, and the resulting current in the antenna produces a THz electromagnetic wave. This radiation is collected and collimated with an appropriate optical system to produce a beam.

Terahertz electronic devices

Metal oxide semiconductor (MOS) transistor is the building block of integrated circuits (ICs) in electronics and is the engine that powers these. Today's most complex ICs, such as microprocessors, graphics and DSP chips, pack more than 100 million MOS transistors on a single chip. Integration of one billion transistors into a single chip will soon become a reality. The semiconductor industry faces an environment that includes increasing chip complexity, continued cost pressures, increasing environmental regulations and growing concern about energy consumption. The observation that the number of transistors per integrated circuit doubles every 18 to 24 months is well-known to industry analysts and many of the general public. New materials and technologies are needed to support the continuation of Moore's law.

TE holds promise of greatly-expanding and numerous applications in detection of biological and chemical hazardous agents, building and airport security, and explosive detection, as well as in radio astronomy, biology and medicine. Many companies are thinking towards exploiting the commercial applications of TE